

# **DAY-1 INTRODUCTION TO NETWORKING HARDWARE**

**Topic: Overview of Modems,  
Routers, Switches, Access Points  
and Network Cables**

**Course: Diploma in Cyber Security**

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## **Task 1: Typical Home Wi-Fi Setup**

### **Task 1: Typical Home Wi-Fi Setup**

A home network normally starts with an internet connection from the ISP. The modem receives the internet connection and passes it to the router.

The router manages the network and provides internet access to different devices. A switch can be connected to the

router when more wired devices are required. An access point can also be connected to extend Wi-Fi coverage in another area.

## **Basic Connection**

**Internet/ISP**



**Modem**



**Router**



**Switch**



**Desktop PC / Smart TV**

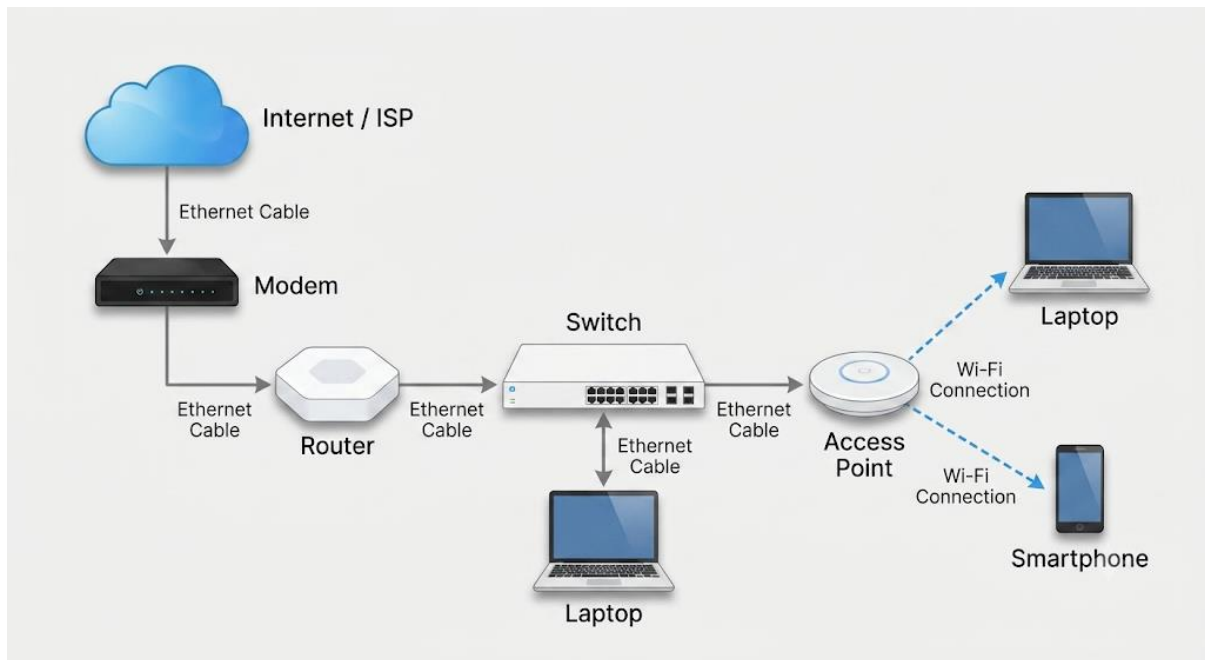
**Router**



**Access Point**



**Laptop / Smartphone**



## Task 2: Functions of Networking Hardware

### 1. Modem

#### Main Function:

A modem connects the local network to the internet service provided by an ISP. It converts the signal received from the ISP into a form that network devices can use.

#### Real-life example:

A modem can be used at home to receive an internet connection from a broadband provider.

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### 2. Router

#### Main Function:

A router connects different networks and sends data to the

correct destination. At home, it is also commonly used to provide Wi-Fi.

**Real-life example:**

A Wi-Fi router is used at home to connect mobile phones, laptops and smart TVs to the internet.

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### **3. Switch**

**Main Function:**

A switch connects multiple wired devices in the same local network and forwards data to the required device.

**Real-life example:**

A switch can be used in an office to connect many desktop computers using Ethernet cables.

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### **4. Access Point**

**Main Function:**

An access point provides wireless network access to devices such as smartphones and laptops. It can also help increase Wi-Fi coverage.

**Real-life example:**

An access point can be installed in another part of a large office or building where the main Wi-Fi signal is weak.

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### **5. Network Cable**

**Main Function:**

A network cable provides a physical connection between networking devices and carries data from one device to another.

**Real-life example:**

An Ethernet cable is commonly used to connect a computer to a router or switch.

## Task 3: Types of Network Cables

### 1. Ethernet Cable

**Description:**

Ethernet cables are commonly used to connect computers, routers, switches and other network devices in a wired network.

**Common Use:**

Used in homes, offices, computer labs and data networks.

**Example:**

RJ45 Ethernet cable connecting a computer to a router.

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### 2. Coaxial Cable

**Description:**

Coaxial cable has a central conductor surrounded by insulation and shielding. It is commonly used for cable television and some broadband connections.

**Common Use:**

Used for cable TV and cable internet connections.

**Example:**

Cable connection from an ISP to a modem.

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### **3. Fiber Optic Cable**

**Description:**

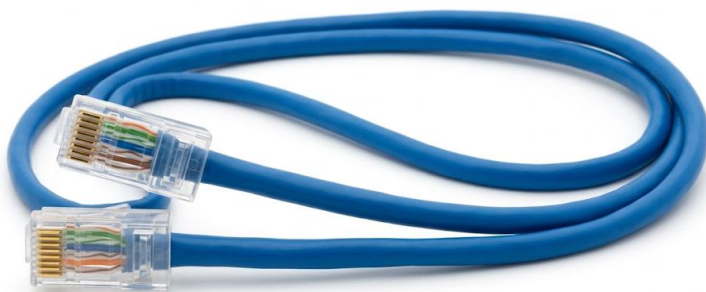
Fiber optic cable uses light to transfer data. It can carry data at very high speeds and is useful for long-distance communication.

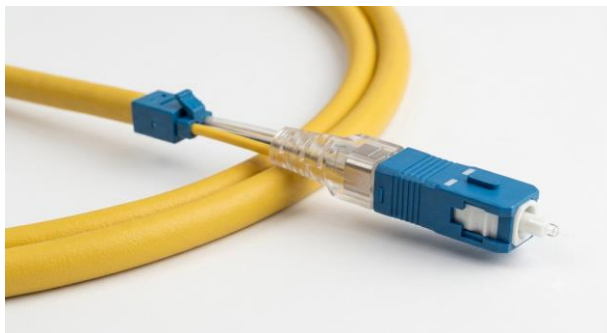
**Common Use:**

Used for high-speed internet, telecom networks and large data networks.

**Example:**

Fiber connection used by an ISP to provide high-speed internet to a building.





## Task 4: Router vs Switch Comparison

| Difference       | Router                                                            | Switch                                                  |
|------------------|-------------------------------------------------------------------|---------------------------------------------------------|
| Main Use         | Connects different networks and provides network/internet routing | Connects multiple devices within the same local network |
| Typical Location | Commonly found at homes, offices and network gateways             | Commonly found in offices, labs and server rooms        |
| Data Handling    | Uses IP addresses to decide where data should go                  | Mainly uses MAC addresses to                            |

| <b>Difference</b>          | <b>Router</b>                                        | <b>Switch</b>                                      |
|----------------------------|------------------------------------------------------|----------------------------------------------------|
|                            |                                                      | forward data inside a LAN                          |
| <b>Internet Connection</b> | Usually connects the local network to the internet   | Normally does not directly provide internet access |
| <b>Connectivity</b>        | Can connect different networks                       | Connects devices within the same network           |
| <b>Wireless Function</b>   | Many home routers also provide Wi-Fi                 | A normal switch mainly provides wired connections  |
| <b>Example</b>             | Home Wi-Fi router connecting devices to the internet | Office switch connecting multiple computers        |

A router and a switch both help devices communicate over a network, but their jobs are different. A router mainly connects different networks, while a switch mainly connects multiple devices within the same local network.



# Conclusion

From this assignment, I learned about the basic networking hardware used in homes, offices and other places.

I understood that a modem is mainly used to connect to the ISP, while a router manages communication between networks and can provide Wi-Fi. A switch is useful for connecting multiple wired devices, and an access point provides wireless network access.

I also learned about Ethernet, coaxial and fiber optic cables and where they are commonly used. The comparison between a router and switch helped me understand that both devices perform different roles in a network.

Overall, this activity gave me a basic understanding of how networking devices work together to create a working network.